



VIT[®]
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

School of Computer Science and Engineering

BCSE305L – Embedded Systems

Module 7.1

Applications of Embedded Systems

Module:7 Applications of Embedded Systems

4 hours

Introduction to embedded system applications using case studies – Role in Agriculture sector, Automotive electronics, Consumer Electronics, Industrial controls, Medical Electronics.

Weather Forecasting

Procedures of Weather forecasting

The weather forecasting has now become a science and it is performed by adopting the following procedures (steps) :

1. recording of weather data (temperature, pressure, wind speed and direction, cloud forms, humidity and precipitation, visibility, storms etc.)
2. collection of weather data from weather recording (observations centers) stations scattered world over including both land and ocean surfaces
3. transmission of weather data collected from major weather stations to sub centers .

4. Compilation of weather data.
 5. Plotting of weather data on maps and daily weather records, synoptic charts etc.
 6. Analysis of weather charts and maps with the help of electronic computers etc.
 7. Final forecasting of weather and numerical modeling .
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Brief information about weather forecasting centre

(source : climatology by Savindra Singh)

Observatories & meteorological station	Ship based observation centre	Radar stations	Upper air weather stations	Three major international centre
9500+	7400+	600+	2300+	Washington D.C. Moscow Melbourne

Tools in weather forecasting

RADIOSONDES

It is an instrument that is carried aloft by a balloon to send back information on atmospheric temperature, pressure and humidity by means a small radio transmitter



Weather satellites



- ▶ The most significant device is weather satellite of different kinds attached with different types of sensors for specific purposes and having varying orbital paths around the earth.
- ▶ The first weather satellite , named TIROS-I was launched in 1960

The sensors which are attached with satellites send back to the images of clouds to earth's centre .

Weather Satellites

(on the basis of orbital paths)

Geosynchronous Satellites

Same rate of movement around the earth as that of the rotational speed of the earth.

These are fixed at the altitude of 35,786 kilometers from the sea level

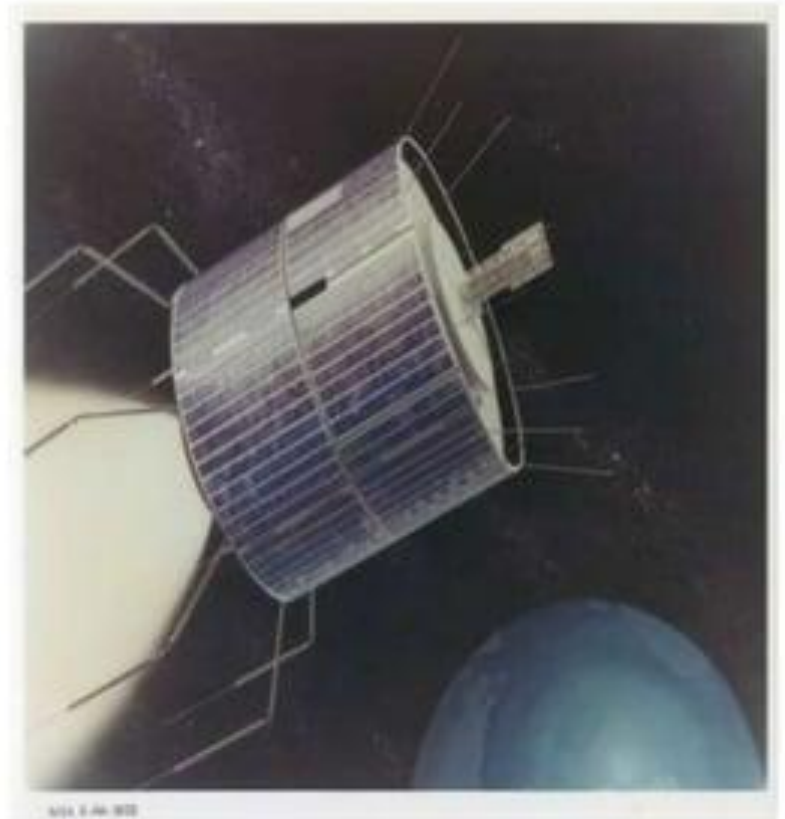
These are called *high altitude satellites*.

Provide scanned pictures of larger areas.

These having *imaging sensors* instruments that provides thermal variations of the atmosphere, moisture etc.

Example: **ATS** series and **GOES** series weather satellites.

GOES & ATS



Polar orbiting satellites

- ▶ These are positioned after their launching by rockets at the latitude ranging between 800 to 1500 kilometers
- ▶ These are also known as low latitude satellites
- ▶ these having *sounding sensors* naming ADVANCED VERY HIGH RESOLUTION RADIOMETER (AVHRR)
- ▶ Example: TIROS and NOAA

NOAA & TIROS



RADAR

- ▶ These are useful tools to obtain detailed information about clouds and storms mainly cyclones , tornadoes, hurricanes.



TYPES OF WEATHER FORECASTING

On the basis of duration of the validity of forecasts into three types as follows :

- 1.Short range weather forecasts
 - 2.Medium range weather forecasts
 3. Long range weather forecasts
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Short-range weather forecasts

- ▶ Most useful predictions of weather wherein forecasts valid from few hours to 48 hours and sometime 72 hours
- ▶ Based on maps weather charts , satellite images
- ▶ Determines change of atmospheric weather of a specific location
- ▶ Persistent method & continuity methods are applied .
- ▶ Helps in transportation and fishermen .

Medium range weather forecasts

- ▶ It covers the time span from 3days to 3 weeks.
- ▶ This prediction is helpful for farmers
- ▶ This prediction is done by calculating the average of past and present weather condition.

Long range weather forecasts

- ▶ It covers a time span from a fortnight to a season of the year.
- ▶ These are basically statements ,exact accuracy is minimum .
- ▶ Not so detailed information.

On the basis of specific purposes for different segments of society into following four types:

- a. Aviation forecasts
 - b. Shipping forecasts
 - c. Local forecasts
 - d. Agricultural-based forecasts
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Source

- ▶ Climatology ; Savindra Singh
 - ▶ Climatology ;D.S.Lal
 - ▶ Wikipedia
 - ▶ Google images
 - ▶ Microsoft office 2007
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Weather forecasting –How it used to be done

- Traditionally weather forecasting relied upon the use of large numbers of different places in the country sending in regular reports to the government's Metrological Office.
- These inputs were recorded and collated, and then used to predict the future weather.

Weather Data:

- The weather data that is collected every 24 hours includes:
 - ☐ Wind direction.
 - ☐ Wind strength.
 - ☐ Maximum and minimum temperature.
 - ☐ Relative humidity.
 - ☐ Number of hours of sunshine.
 - ☐ Precipitation (Rainfall/Snow).

Problems

- How can data be collected regularly? (i.e. Every hour, on every day, throughout the whole year)
- How can human error be avoided?
- How can more Frequent measurements.
- How accuracy of readings (sometimes the instruments are read incorrectly)

■ **Data Logging (Answer)**

- which can be automated collects data over a certain period of time and does not require any human intervention.
- Since there is no human errors, the measurements will be accurate.

Data Logging (Answer)



- which can be automated collects data over a certain period of time and does not require any human intervention.
- Data logging stands for collection and recording of information.
- It can be manual, where you have to take each reading yourself or automated, where you get a computer or machine to take readings as often as you choose.
- Data logging normally makes use of sensors - these are devices that take measurements and feed the data back to the computer.
- Example:
 - In hospital (ICU)
 - Racing Cars



Data logging devices

- Temperature sensors.
- Wind speed sensors.
- Wind direction sensors.
- Rainfall detectors.
- Light detectors.
- Humidity sensors.



Automated Weather Station

1. Temperature sensors

- These are heat-sensitive sensors that produce an analogue temperature signal which is converted (via an analogue-to-digital converter) to a digital signal.
- This signal is then stored in a microprocessor that is downloaded regularly.



2. Wind speed sensors

- A revolving anemometer (which it spins faster or slower depending upon the speed of the wind) is used to measure wind speed.
- An optical sensor counts the number of times the anemometer revolves in a given length of time, and converts the number into a binary digital signal that can be stored and download later.



3. Wind direction sensors

- These use a grey code disk attached to a weather vane.
- As the weather vane moves, optical sensors read the disk and generate a three bit binary pattern that can be stored for later downloading.

4. Rainfall detectors

- Rainfall is collected in small buckets which, when full, tilt and empty.
- An optical sensor detects each time a bucket tips, and saves the number of 'tips' as a digital number that can be downloaded later.
- This measures how much rain has fallen over a period of time.



5. Light detectors

- These use a special diode that registers the number of times and the length of time the sun shines during a given length of time.
- This analogue information is converted into digital signal that can be stored and later downloaded
- Digital signal used by microprocessor to determine sunrise and sunset

Analogue –to- digital

Analogue

- Which is continuously variable and do not jump in steps from value to value.
- Analogue need to be converted to digital values using an ADC.
- All quantities measured by the weather reporter.
- A rainfall meter measures rainfall in this system is analogue.
- Temp don't jump from one degree straight to the next, there are many values in between.

Digital

- That jump from one value to the next.
- It's a discrete values.
- Can be fed directly into the processor
- Most computers process only digital values.
- The sensors itself counts the number of buckets that are filled.
- Wind speed anemometer measures wind speed is digital value.

Advantages of data Logging

- Data Logging can be used in remote or dangerous situations
- Data logging can be carried out 24 hours a day, 365 days of the year
- Time intervals for collecting data can be very frequent and regular, for example, hundreds of measurements per second
- can be set up to start at a time in the future
- No need to have a person present
- Data logging is often more accurate because there is no likelihood of human error.
- Data logging devices can be sent to places that humans can not easily get to. e.g. to the planet Mars or onto a roof of a tall building to get to a weather station.
- Graphs and tables of results can be produced automatically by the data logging software.

Disadvantages of data Logging

- If the data logging equipment breaks down or malfunctions, some data could be lost or not recorded
- Equipment can be expensive for small tasks.
- The equipment will only take readings at the logging interval which has been set up.
- If something unexpected happens between recordings, the data will not be collected.

Weather Satellites

- By using weather reporter and images from the satellite orbiting the earth.
- **How weather satellite produce pictures:**
- Transmit pictures as coded radio signals.
- The coded signal is picked up with satellite dish on the earth.
- **Use of satellite images**
- Determine the amount of cloud covered.
- Which direction the wind is likely to come, we can predict the weather.
- Relative temperature of land and see with the help of infra red.
- Still pictures can be taken over a period of time and stored.



Weather Monitoring System



Problem Statement:-

- Weather conditions have an impact on human activity, and monitoring the weather may aid in regulating it. It is crucial to observe and research the local weather patterns.
 - User has few options for learning weather information, such as temperature, humidity, and wind speed. Without a weather station, a user cannot get notifications of heat waves, severe winds, or other weather-related emergencies.
 - Additionally, it is challenging to make weather forecasts without data. When using a weather station, a user may also access historical data. The user can identify measurement trends. This will make it possible for users to examine trends more successfully.
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Objective:-

- To create a real-time weather station that enables users to obtain data instantly and anywhere.
- Using the Internet of Things, create a weather station.
- To create data for the user and assess the capabilities and efficacy of the weather station.



Scope:-



gathers information on air pressure, humidity, and temperature from any station.



Every five minutes, the weather station will send data to the database.



Real-time data is saved and delivered to a local server.

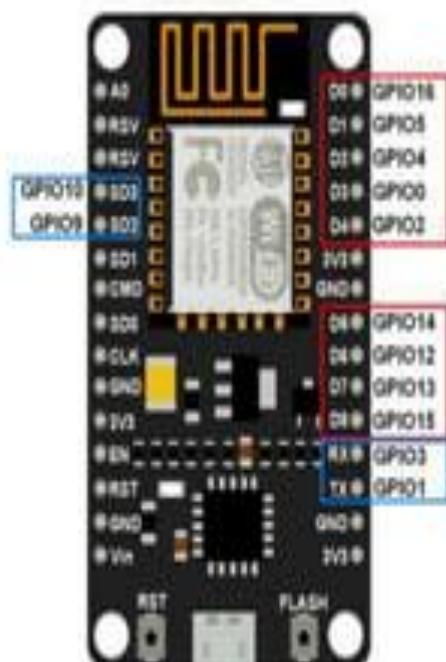
Materials Required:-

1. NodeMCU
2. DHT11 Sensor
3. Rain Sensor
4. LDR sensor
5. 16x2 LCD Display
6. Blynk app
7. Arduino IDE
8. Jumper Wires.



1. NodeMCU

- It is an open source, single-board microcontroller-based IOT platform that is inexpensive. It has a storage capacity of 4MB and 128kB of memory. A dual in-line package (DIP) is a circuit board that combines a USB controller with a smaller surface-mounted board that houses the MCU and antenna. It originally came with firmware that works on the ESP8266 Wi-Fi SoC Systems. Access to the GPIO (General Purpose Input/Output) is also provided.



DHT11 Sensor



1 - Vcc
2 - Data/out
3 - Ground



A crucial, extremely low-effort computerized temperature and humidity sensor is the DHT11.



It measures the air around it using a thermistor and a capacitive humidity sensor, and it outputs digital data on the data pin.



The digital signal output of the DHT11 Temperature and Humidity Sensor is available.



It has an 8-bit high-performance microprocessor built in.



High dependability and exceptional long-term stability are guaranteed by its technology. It is of outstanding quality, responds quickly, is interference-free, and performs

DHT 11 Sensor Specifications:-

- Operating Voltage: 3.5V to 5.5V
- Operating current: 0.3mA (measuring) 60uA (standby)
- Temperature Range: 0°C to 50°C
- Humidity Range: 20% to 90%
- Resolution: Temperature and Humidity both are 16-bit
- Accuracy: +1°C and +1%



Rain Sensor

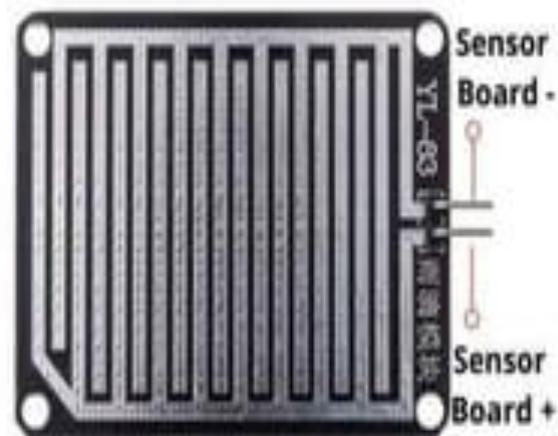
A simple instrument for detecting rain is the rain sensor module. When a raindrop passes through the rainy board, it may be utilized as a switch, and it can also be used to gauge how hard the rain is falling.

The analogue output is utilized to monitor decreases in rainfall volume. The LED will turn on when the induction board has no raindrops and the DO output is high when connected to a 5V power supply.

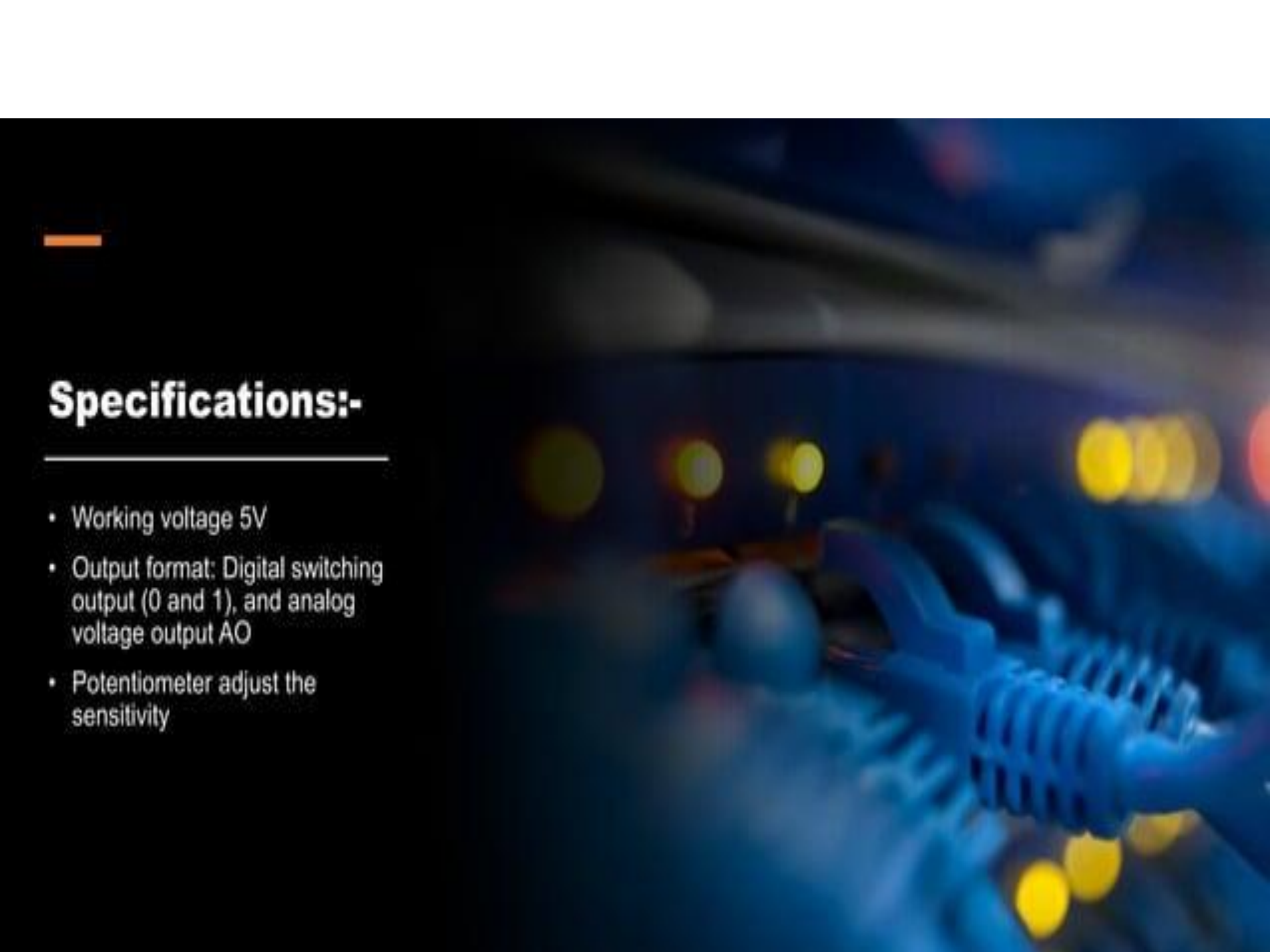
When the DO output is low and a little amount of water is dropped, the switch's indicator will turn on.

Remove the water droplets, and the output is high when it is returned to its original state. A switching device that is triggered by rainfall is known as a rain sensor or rain switch.

DO = Digital Output
AO = Analog Output



Sensistivity Adjustment



Specifications:-

- Working voltage 5V
- Output format: Digital switching output (0 and 1), and analog voltage output AO
- Potentiometer adjust the sensitivity

LDR Sensor



Essentially, it is a variable resistor, where the resistance changes as the light intensity increases. More light means less resistance. Light intensity and resistivity are inversely related.



The IR sensor and LDR sensor interfaces for Arduino are identical. If you wish to interface it, you also have two alternatives. Both the output digital pin and the analogue pin are used in one.

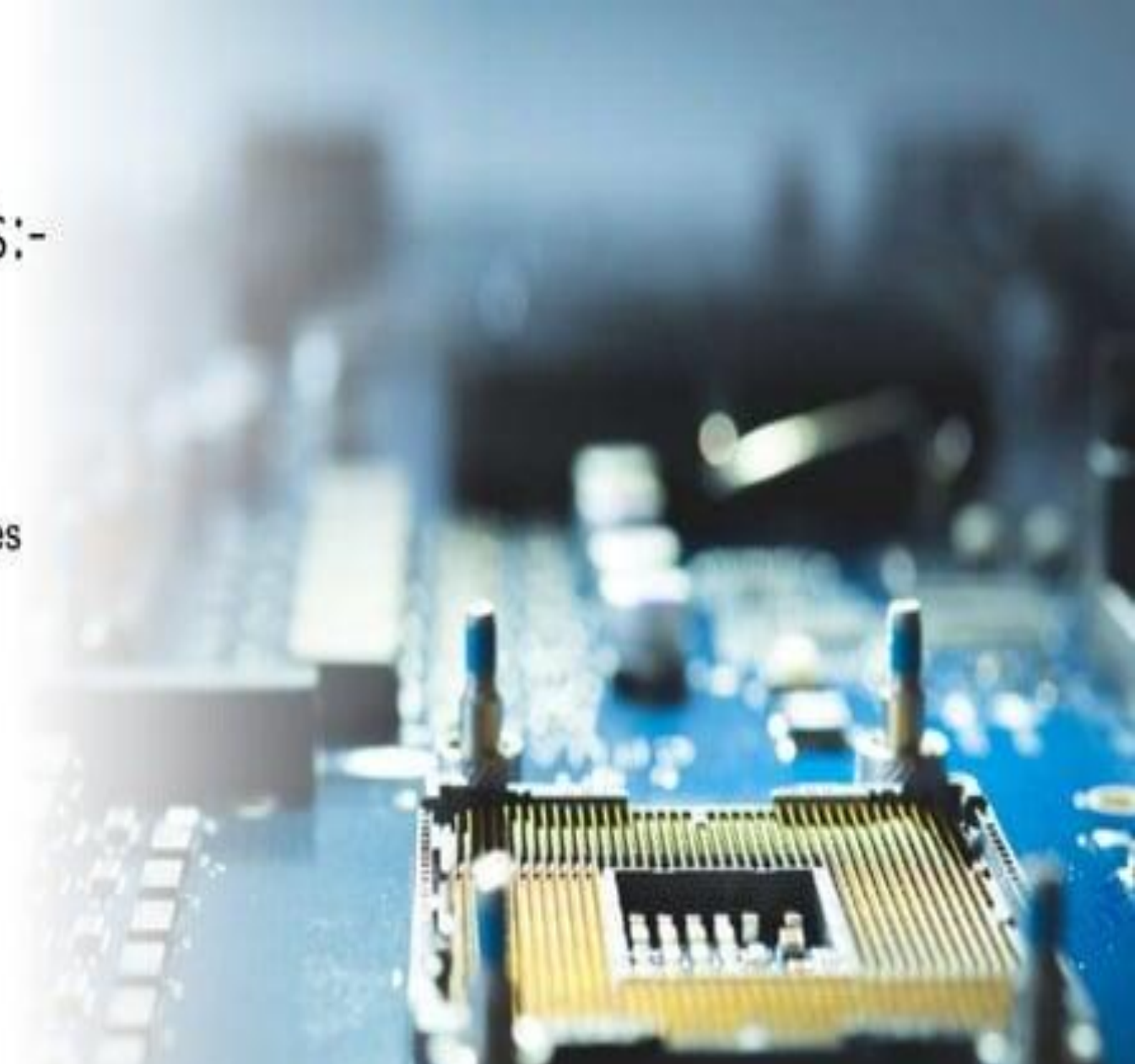


LDR sensor to distinguish between day and night. The hardware LCD's light will be out during the day, and on the Blynk App, it will display "Red light off." But at night, the LCD backlight will activate, just like on the Blynk App.



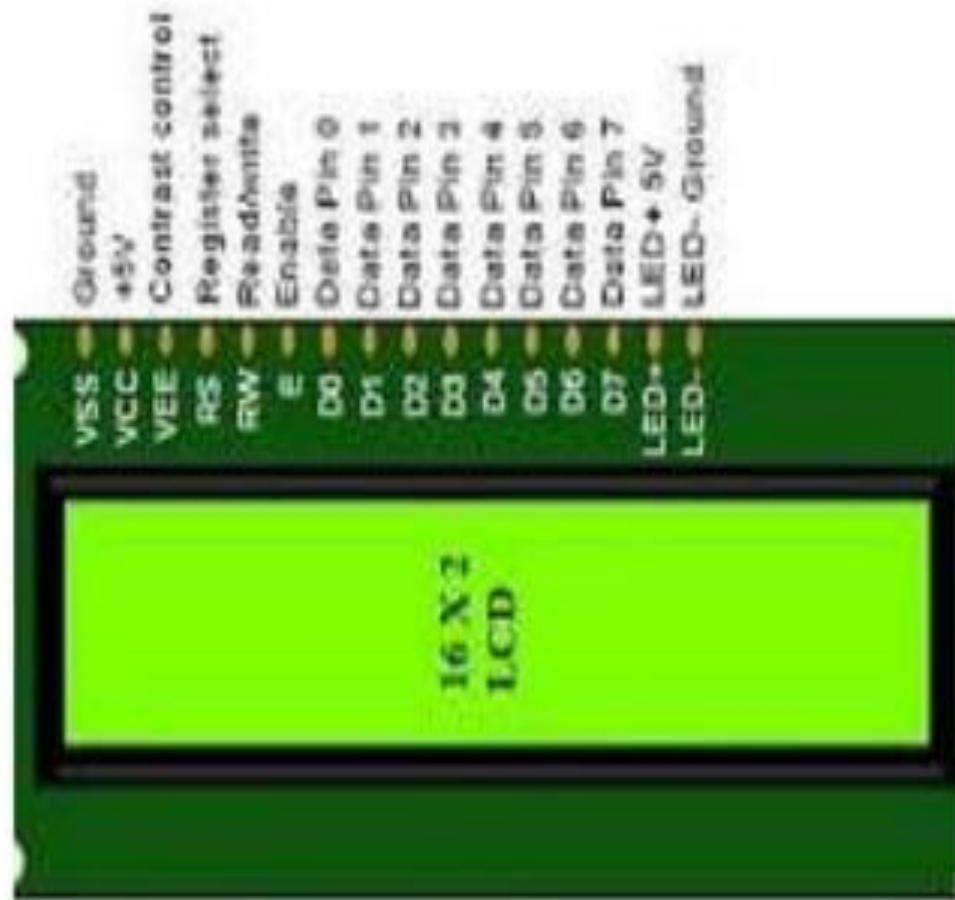
Specifications:-

- Sensitivity is High
- Simple & Small devices
- Easily used
- Inexpensive



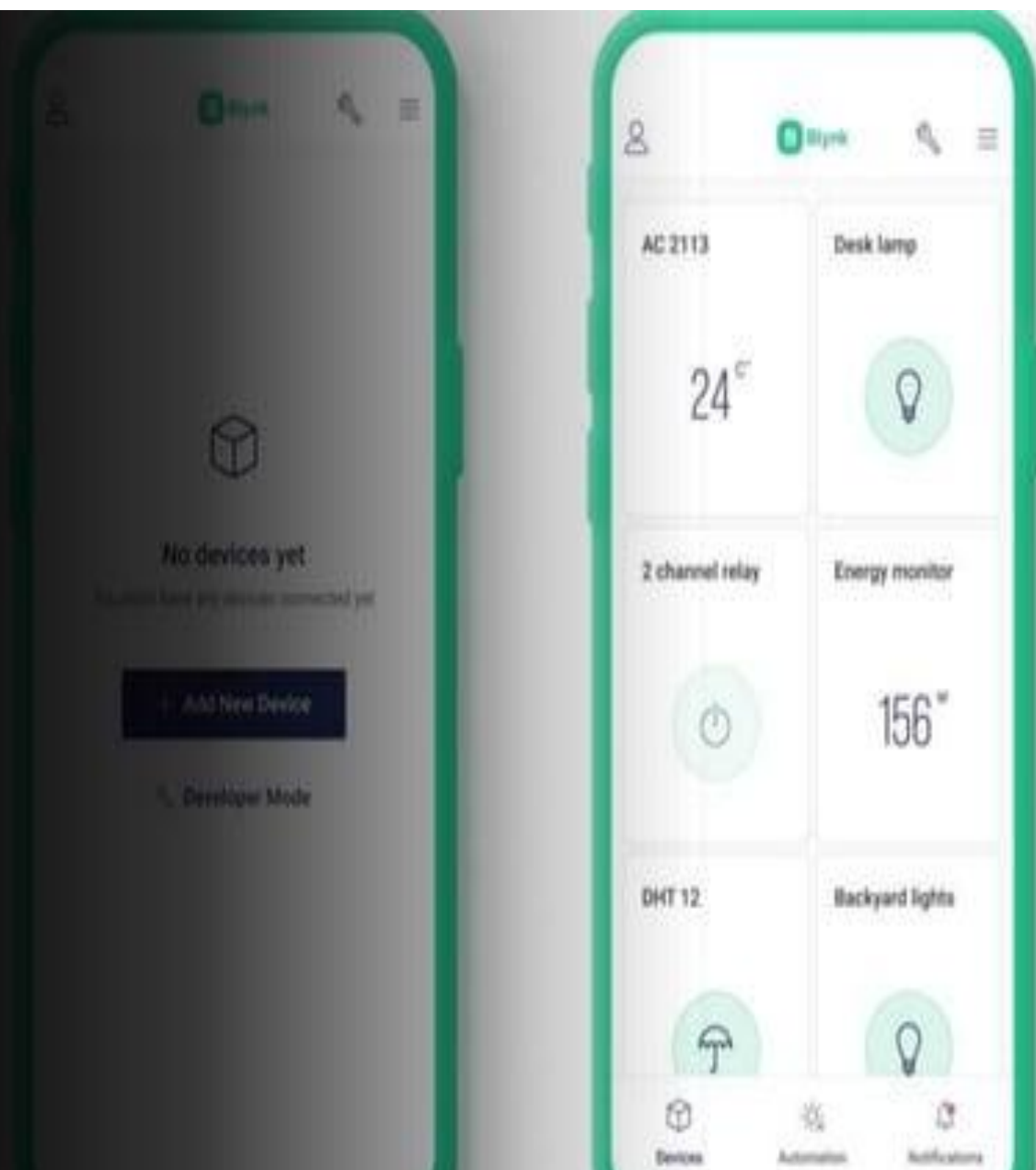
16X2 LCD Display

- A liquid crystal display, sometimes known as an LCD, is essentially a display device made with liquid crystal technology. When creating projects based on real-world electronics, we require a way to show output values and messages.



Blynk App

- Blynk was used to collect all of the data. The user must first register in order to create an account.
- The Auth Token will then be sent by Blynk to the registered email address.
- The user may utilize the application and create their own data, such as a graph, display value, button, table, and others when the Auth Token has been sent.



Arduino IDE

- The Arduino IDE has specific code organization guidelines to support the languages C and C++. A software library from the Wiring project, which offers several standard input and output operations, is provided by the Arduino IDE. The NodeMCU is loaded into the Arduino IDE via a loader programme in the board's firmware using the programme that the Arduino IDE uses to transform the executable code into a text file in hexadecimal encoding.





Advantages:-

- It doesn't need any assistance from people.
- Weather alerts are available in advance.
- In this method, costs are cheap and efforts are reduced.
- There is great accuracy.
- Self Protection.
- Environment monitoring in a clever approach efficient.
- The weather monitoring system will play a very crucial role in the field of industry, farms, agriculture Etc.

Circuit Diagram

$$F = G \frac{m_1 m_2}{d^2}$$

$$i\hbar \frac{\partial}{\partial t} \psi = \hat{H} \psi$$

$$\phi(x) = \frac{1}{\sqrt{2\pi\sigma}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$F - E + V = 2$$

$$E = mc^2$$

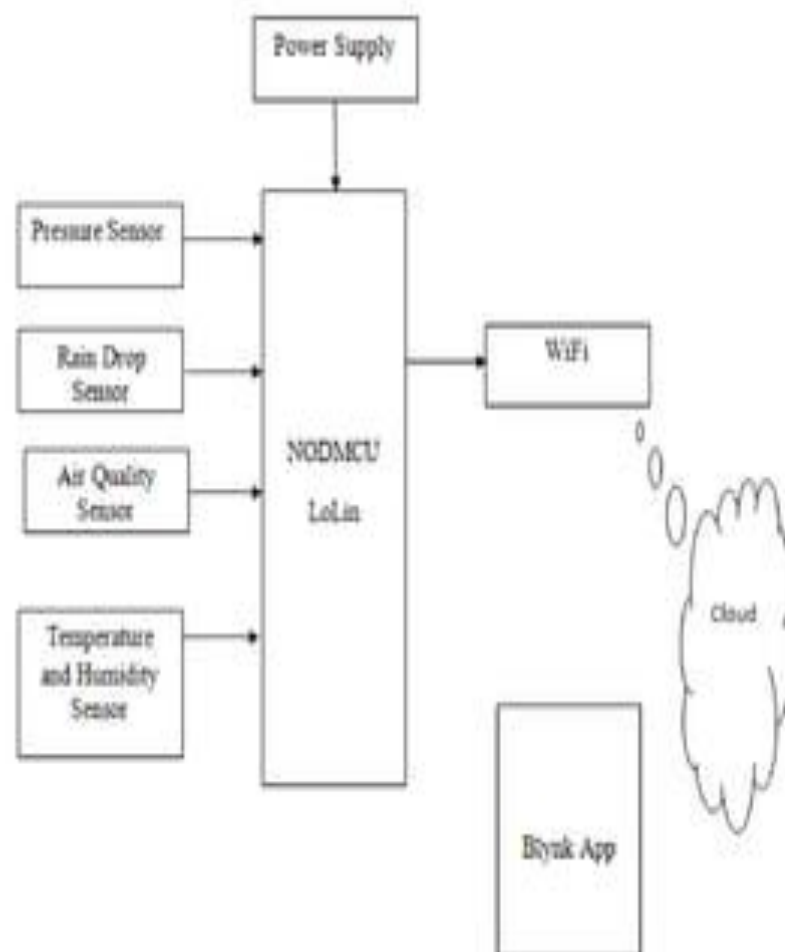
$$ds \geq 0$$

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$

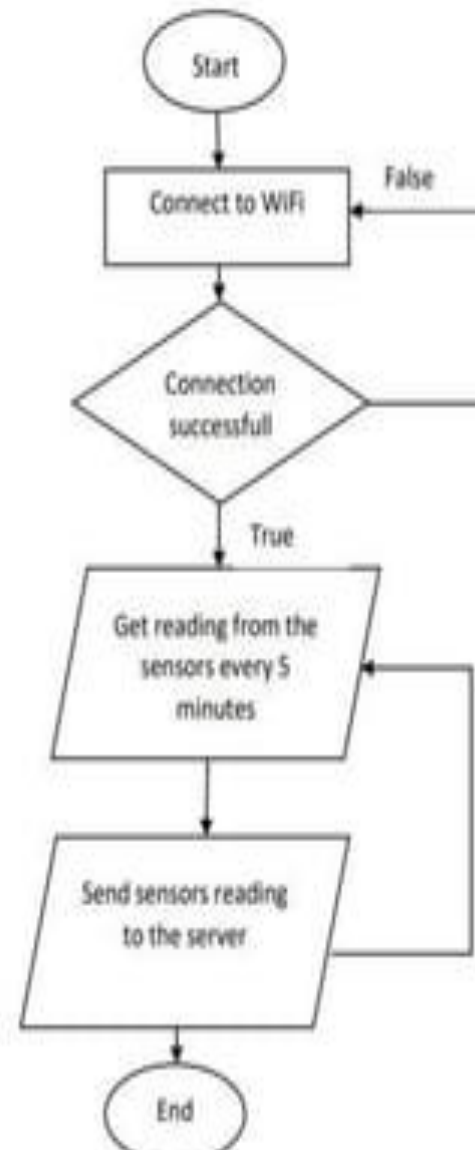
$$\underline{df} = \lim \underline{f(t)}$$



Block Diagram:-



Working model:-



Conclusion:-

- Maintaining a weather station in the environment for monitoring and self-defense (smart environment) enables the environment. The environment's sensor devices must be used for data collection and processing in order to achieve this.
- We can make the environment more realistic by using sensor devices in it. The user will then have access to the gathered data and the analysis' findings over Wi-Fi.
- An effective, low-cost embedded system that monitors the surroundings is provided. Additionally, it uploaded the sensor parameters to the cloud. Future analyses will benefit from this data, which can also easily shared with other users.

Potential of VR in agriculture

VR can solve these problems in a jiffy if its applications are brought in agriculture. VR is cheaper than most of the current agricultural technologies. Furthermore, it doesn't need a huge technical know-how of its usage. Its simple user interface and great immersive technology is an additional advantage. Overall, VR in agriculture is promising and could improve its practices manifold.

Let us take a look at some of the applications VR can have in future:

Virtual Crops

VR can help visualise crop patterns into future. Additionally, it could show crop growth based on weather patterns, practices etc. Researchers can collect data, test and observe the crop growth and suggest changes in practices. This will help farmers prepare their crops against any unwelcome environmental changes.

Virtual crop monitoring:

Drones having 360 degree camera can give virtual tours of the crop area. Immersive technology of VR could help farmers monitor the crop. Furthermore they could check damages, keep a tab on the state of crop growth etc.

Equipment check-ups

Some equipment are complex and dangerous to handle. This makes it difficult for regular maintenance and surveying. Hence, VR could help realise the working of the equipment, and to give an immersive view of its condition and smaller components.

Weather forecast

VR can step up the weather forecast by giving weather conditions that could stimulate or stunt the crop growth. With satellite technology and readings, farmers and researchers could forecast droughts or floods alongside their effects on the land.

Checks on livestock

Farmers maintaining livestock over large areas could also benefit from VR. They can keep a check on the health of the livestock, as well as keep a count or see for looming threats. This will benefit a whole different industry in general. Furthermore, VR equipment don't cost much thus making it more profitable option. There are various other applications which VR could have in agriculture. Agriculture is a huge field, and with increasing potential and development of VR we are going to see a major technological upheaval in the near future.

<https://rawe2020.in/wp-content/uploads/2022/10/final-report.pdf>

<https://www.slideshare.net/RedmondRaminShamshir/digital-agriculture-virtual-reality-agricultural-robotics>